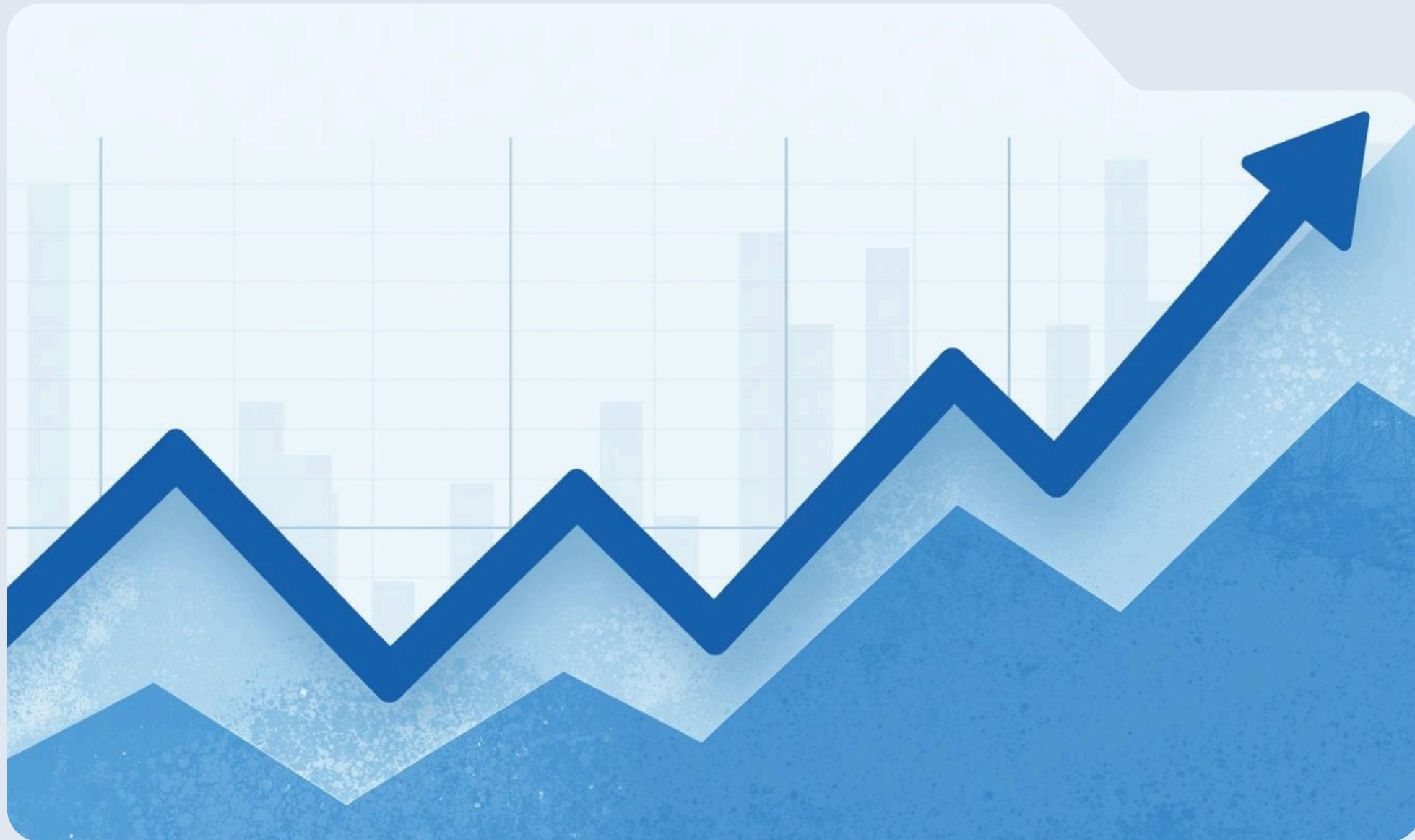
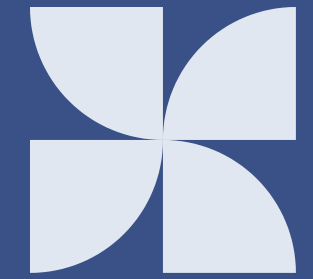


Expected Shortfall and Value-at-Risk Scaling



Presented by

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T7 work projet

Abstract

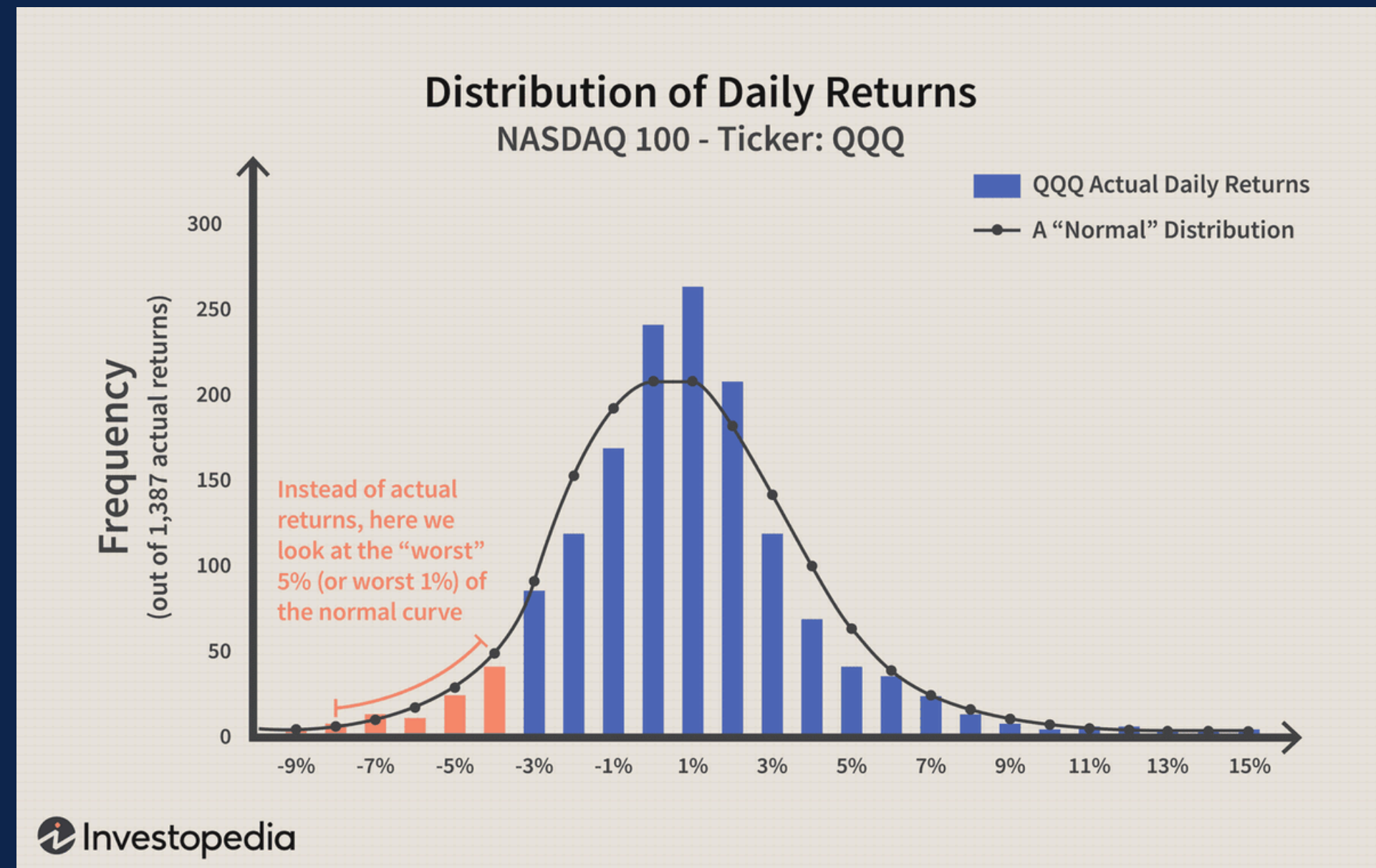
This study aims to **compare scaling methods** for Value-at-Risk (VaR) and Expected Shortfall (ES) specifically addressing high confidence levels and time horizons, incorporating adjustments for non-normality and autocorrelation in profit-and-loss data to enhance risk management practices. The choice of this term project conducted at work has been driven by a wish to use some of the skills learned over the past two years (R, SQL, Python, Time Series, and Data Exploration and Vizualisation).



Risk Measures Overview

Understanding VaR and Expected Shortfall

Value-at-Risk (VaR) quantifies potential loss at a specific confidence level, while Expected Shortfall (ES) provides a deeper insight by averaging losses beyond the VaR threshold. The latter is more useful to explore extreme scenarios, however regulatory constraints also push to estimate the former adequately.



Challenges in Scaling

Addressing Key Issues in Risk Measurement

Non-Gaussian PnL

Non-Gaussian profit-and-loss distributions introduce significant **challenges** in accurately assessing potential losses and risks in financial markets.

Autocorrelation

Positive autocorrelation in overlapping windows can lead to **under or overestimation** of risk, impacting the reliability of traditional scaling methods.

Overestimation of Risk

The commonly used $\sqrt{\text{time}}$ scaling method often results in **overestimating** risk for longer horizons, leading to inadequate capital requirements.

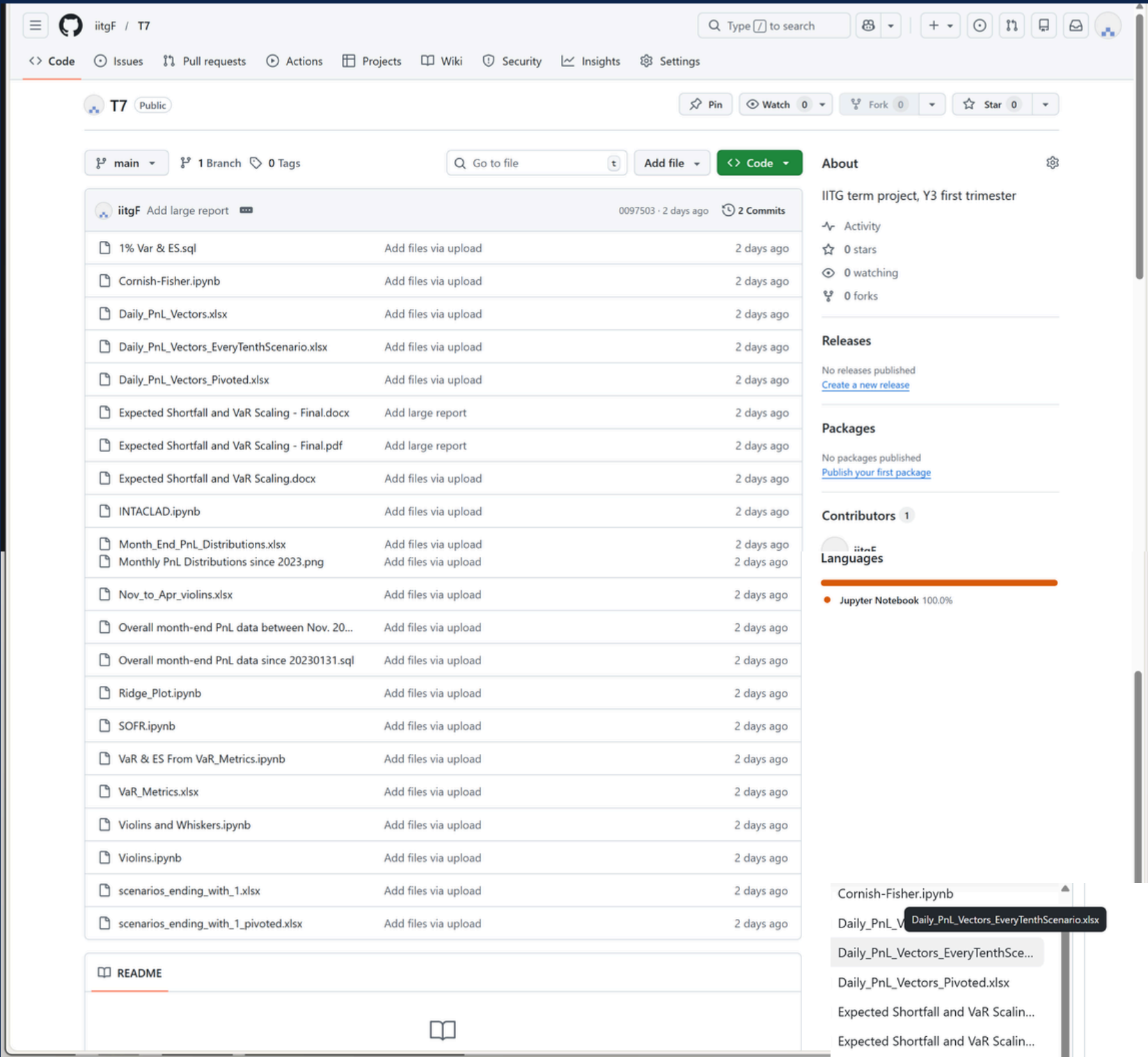
Methodology Overview

Data and Analytical Approaches

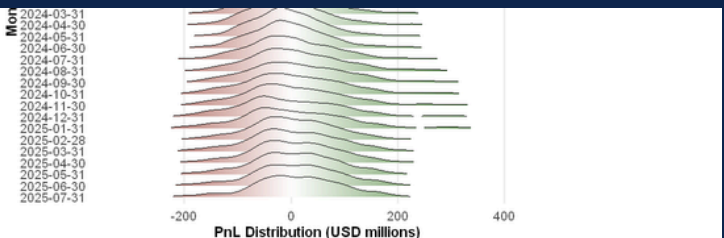
This study utilizes a dataset of 10-day historical PnL from January 2023 to July 2025, employing methods such as historical simulation and normal estimation for risk analysis.

Databases were queried using SQL, calculations and estimations were made using Python, and charts using R.

All code is available on GitHub and will be presented during the Viva Voce.



- Cornish-Fisher.ipynb
- Daily_PnL_Vectors_EveryTenthScenario.xlsx
- Daily_PnL_Vectors_EveryTenthScenario.xlsx
- Daily_PnL_Vectors_Pivoted.xlsx
- Expected Shortfall and VaR Scaling - Final.docx
- Expected Shortfall and VaR Scaling - Final.pdf
- Expected Shortfall and VaR Scaling.docx
- INTACLAD.ipynb
- Month_End_PnL_Distributions.xlsx
- Monthly PnL Distributions since 2023.png
- Nov_to_Apr_violins.xlsx
- Overall month-end PnL data between Nov. 2023 and Nov. 2024.xlsx
- Overall month-end PnL data since 20230131.sql
- Ridge_Plot.ipynb
- SOFR.ipynb
- VaR & ES From VaR_Metrics.ipynb
- VaR_Metrics.xlsx
- Violins and Whiskers.ipynb
- Violins.ipynb
- scenarios_ending_with_1.xlsx
- scenarios_ending_with_1_pivoted.xlsx



Cornish-Fisher Expansion

```
In [6]: # Only run this ONCE as needed, not each time you run your script
# install.packages("moments")

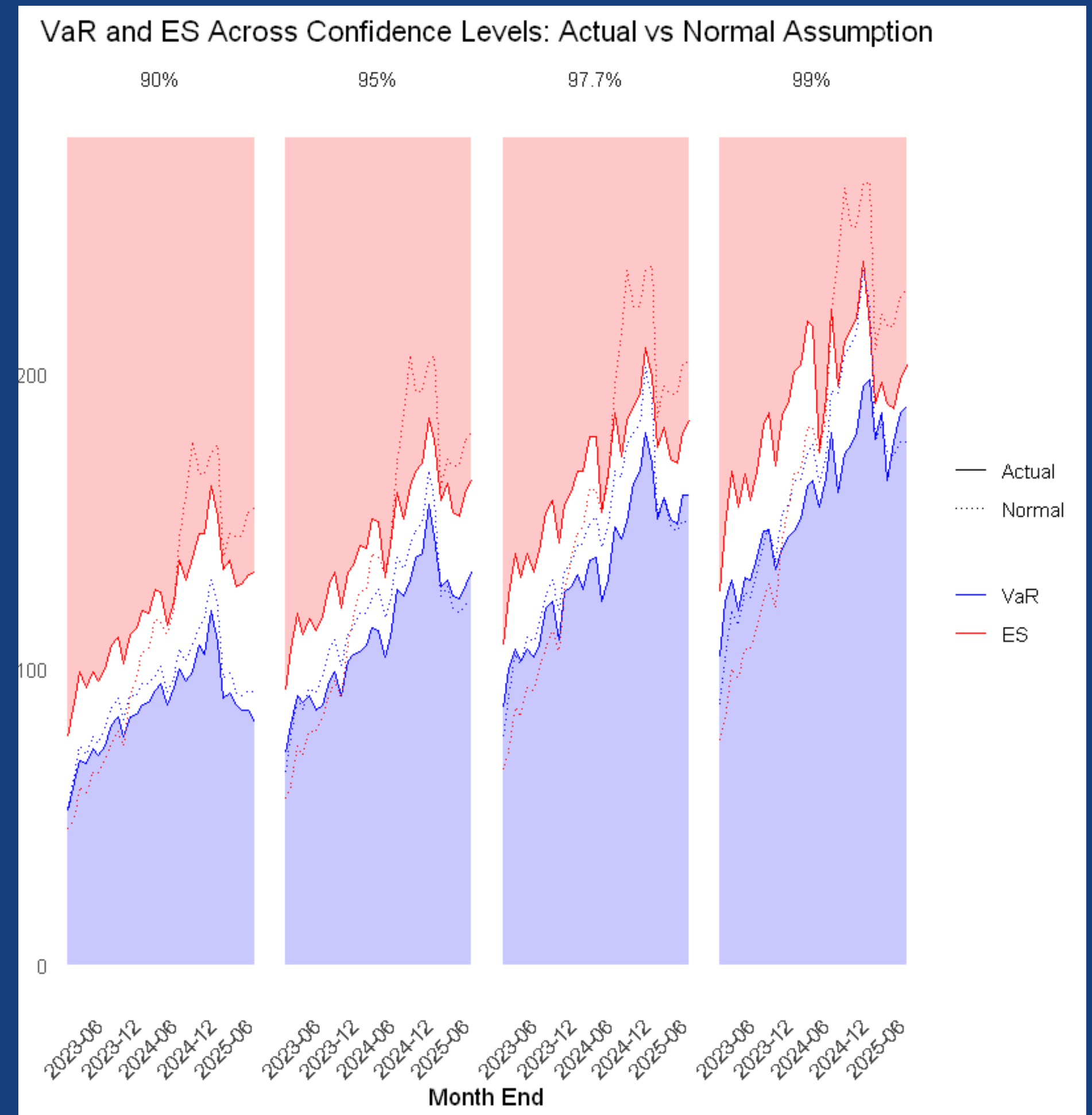
library(readr)
library(dplyr)
library(moments) # for skewness and kurtosis
library(openxlsx) # for Excel export

# Cornish-Fisher VaR function
cornish_fisher_var <- function(pnl_series, alpha = 0.01) {
  mean_val <- mean(pnl_series)
  std_val <- sd(pnl_series)
  s <- skewness(pnl_series)
  k <- kurtosis(pnl_series) - 3 # moments::kurtosis gives excess by default
  z <- qnorm(alpha)
  z_cf <- z +
    (z^2 - 1) * s / 6 +
    (z^3 - 3*z) * k / 24 -
    (2 * z^3 - 5 * z) * s^2 / 36
  var_cf <- -(mean_val + z_cf * std_val)
```

Empirical Results

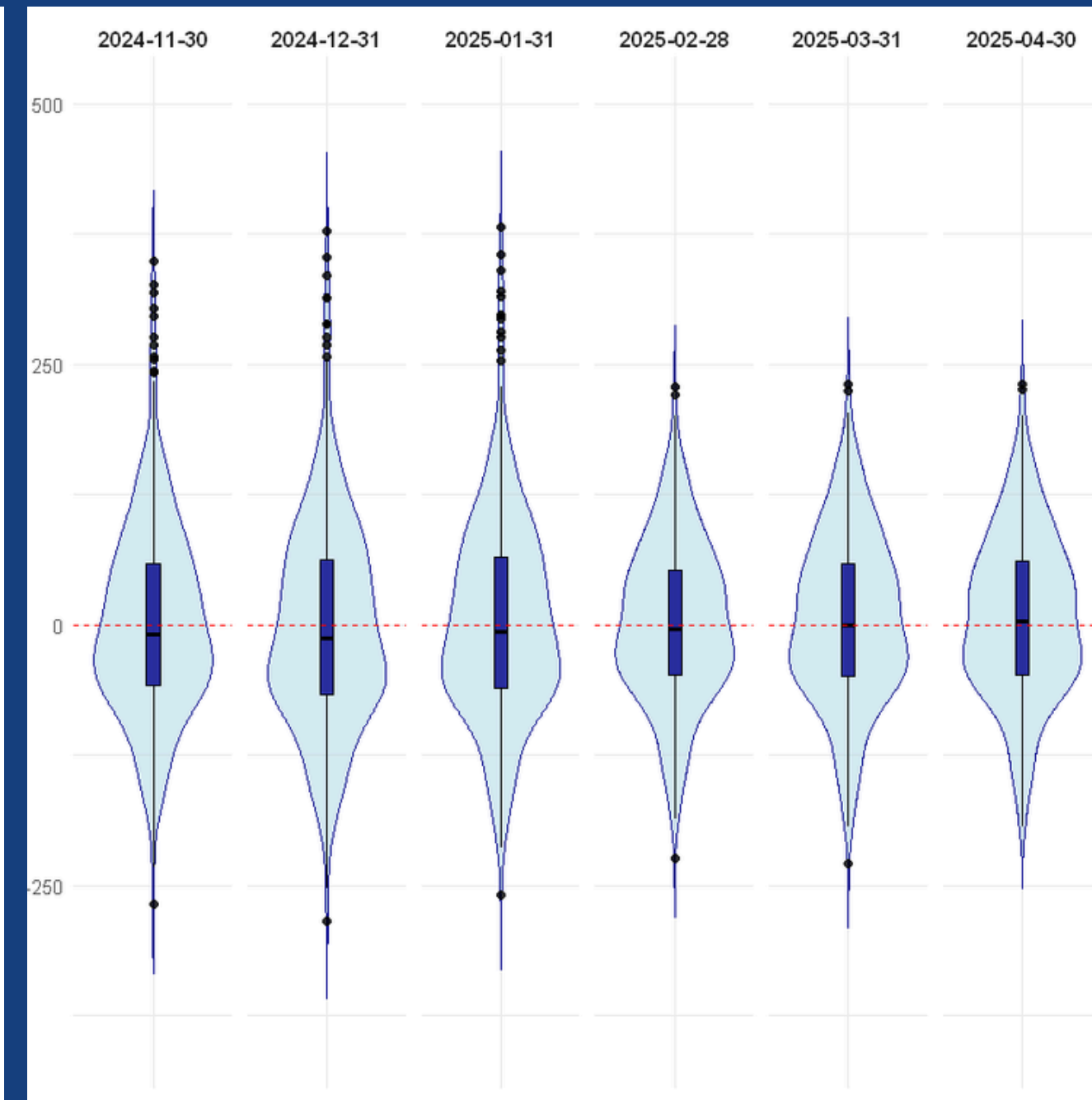
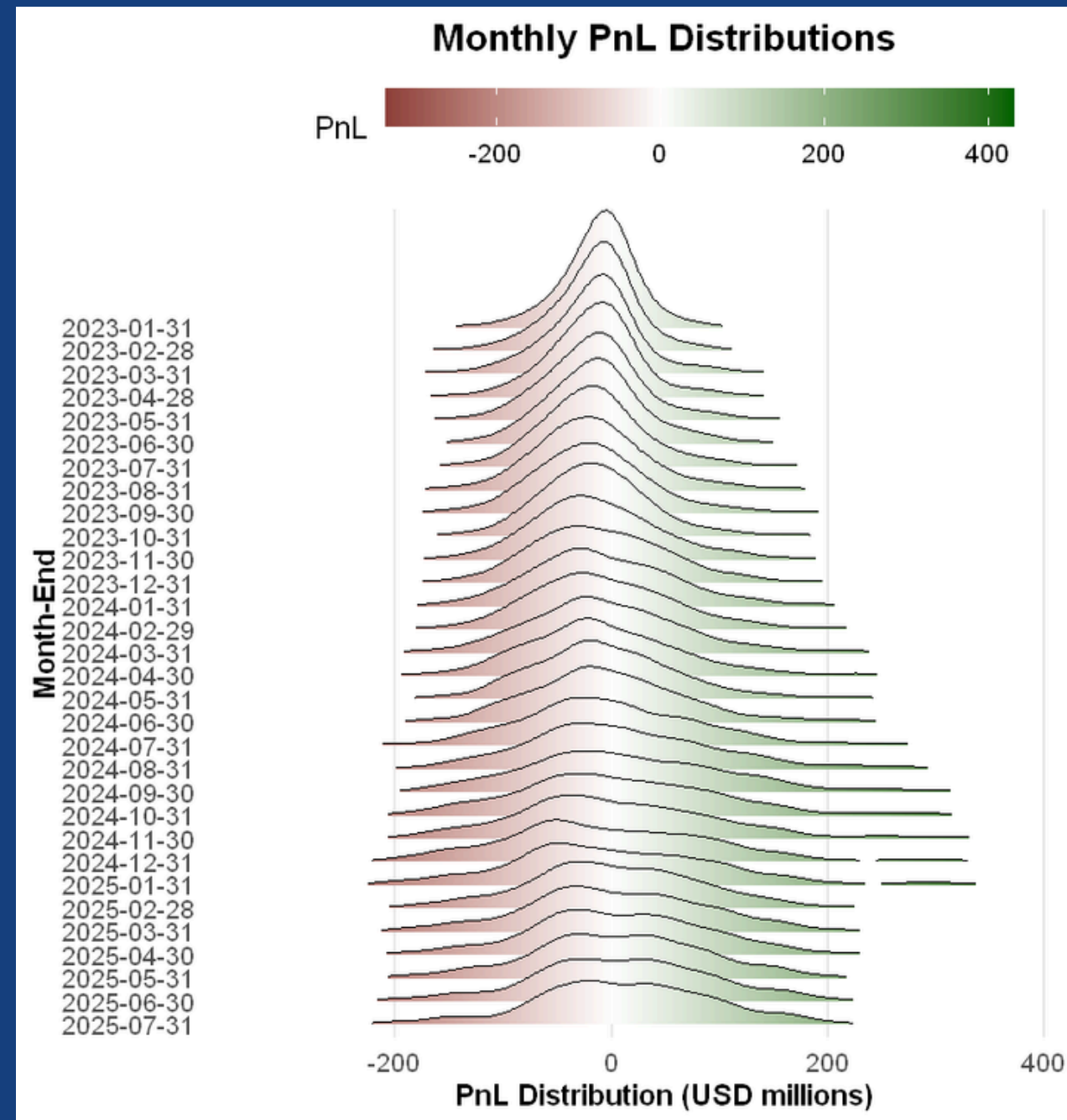
VaR and ES Comparison Insights

This section examines the differences between historical and normal Value-at-Risk (VaR) and Expected Shortfall (ES) measures across various confidence levels, highlighting the consistent tail risk sensitivity of ES. Divergence between historical and normal estimates of Value-at-Risk (VaR) and Expected Shortfall (ES) grow at higher confidence levels. The increasing gap highlights the importance of refine the use of historical data for more accurate risk assessments.



Non-Normality

The presence of **fat tails** and skewness significantly impacts risk measures. The Cornish–Fisher expansion efficiently adjusts quantiles for these characteristics, enhancing the accuracy of estimations under non-normal conditions. This method better reflects actual market behavior during volatile periods, improving risk assessments.



Cornish–Fisher Adjustment

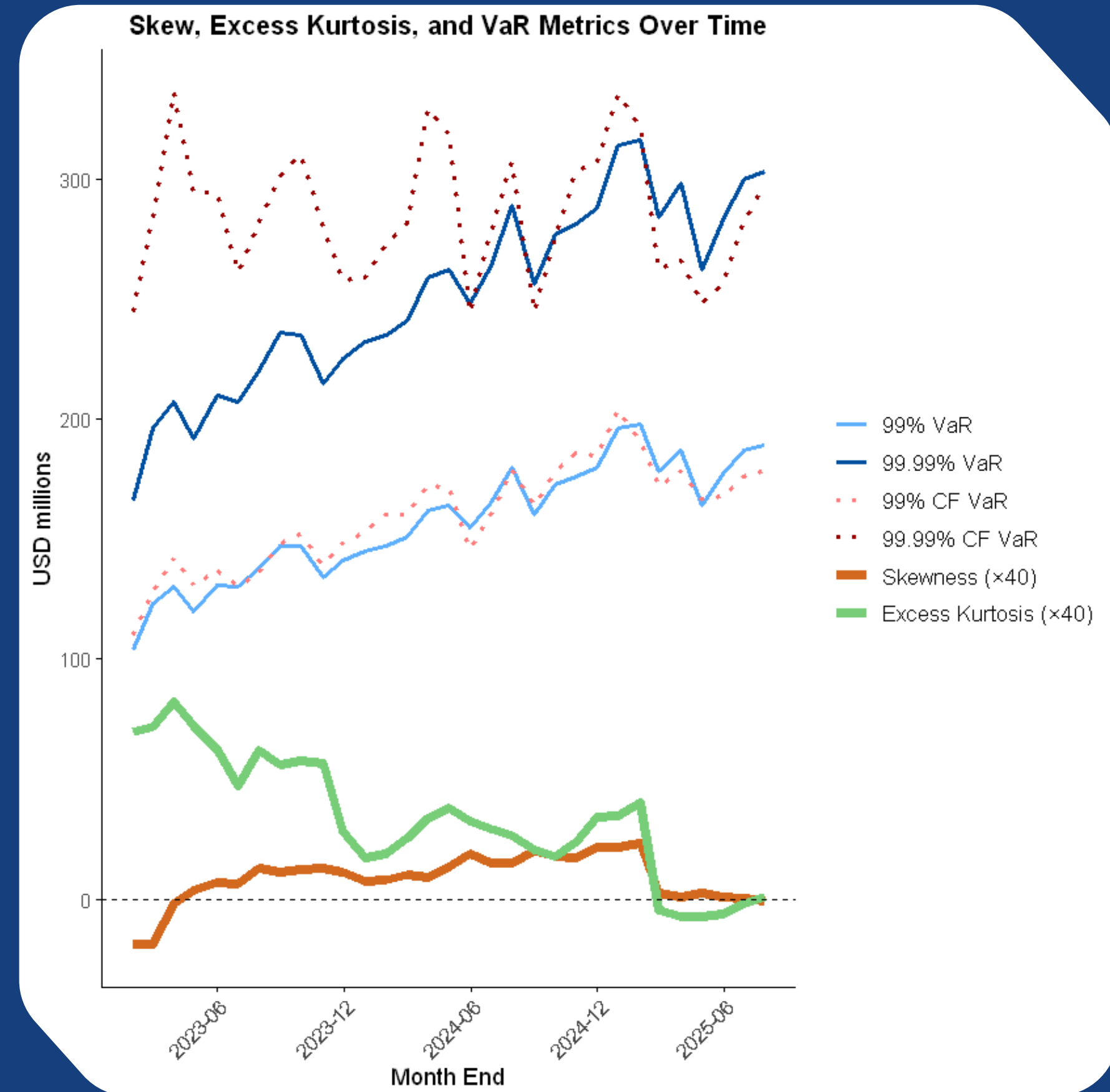
Enhancing Risk Estimates through Techniques

The Cornish–Fisher expansion adjusts quantiles for **skewness** and **kurtosis**, improving estimates in non-normal distributions; this approach enhances accuracy in risk assessment for extreme events.

$$Z \approx z + \frac{S}{6}(z^2 - 1) + \frac{K}{24}(z^3 - 3z) - \frac{K^2}{36}(2z^3 - 5z)$$

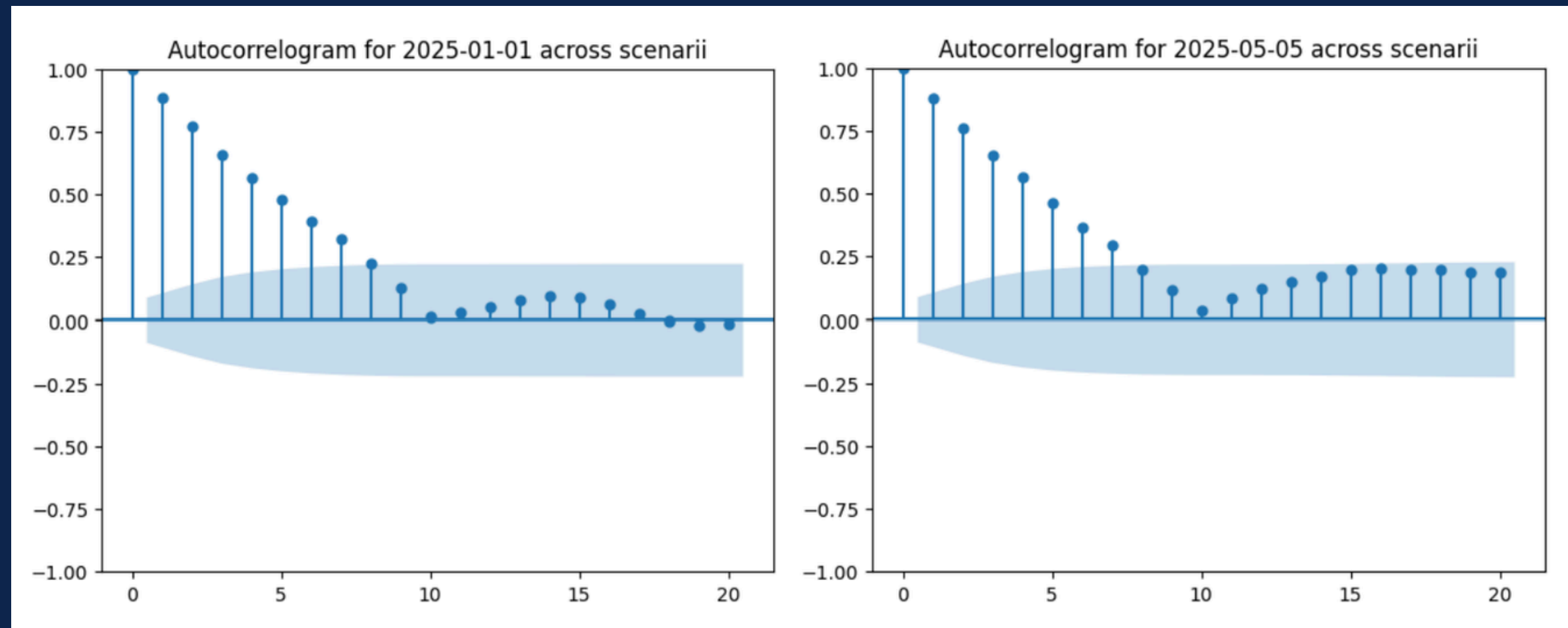
where:

- z is the standard normal quantile;
- S is skewness (third order moment), and K the kurtosis (fourth order moment).



Autocorrelation Effects

Autocorrelation, with a coefficient of approximately **0.5**, indicates significant dependence in overlapping profit-and-loss (PnL) windows. Traditional scaling methods, relying on $\sqrt{\text{time}}$ adjustments, tend to **overestimate risk**; our adjusted scaling factor of about **1.67** provides a more accurate reflection of long-term risk estimates.



Conclusions

Expected Shortfall (ES) is superior for capturing tail risk, but its estimation at extreme quantiles remains challenging. Traditional scaling methods are unreliable due to non-normality and autocorrelation.

Enhanced scaling techniques such as the Cornish-Fisher expansion and autocorrelation analysis can substantially improve VaR assessment accuracy in financial contexts.

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